

BOOKS:

```
mysql> use bookflow_db;
Database changed
mysql> CREATE TABLE Members (member_id INT PRIMARY KEY, full_name VARCHAR(100), email VARCHAR(100) UNIQUE);
ERROR 1050 (42S01): Table 'members' already exists
mysql> CREATE TABLE Loans (loan_id INT PRIMARY KEY, member_id INT, book_id INT, loan_date DATE, FOREIGN KEY (member_id)
REFERENCES Members(member_id), FOREIGN KEY (book_id) REFERENCES Books(book_id));
Query OK, 0 rows affected (0.06 sec)

mysql> CREATE TABLE Donation_History (donation_id INT PRIMARY KEY, book_id INT, donor_name VARCHAR(100), donation_date DATE, FOREIGN KEY (book_id) REFERENCES Books(book_id));
Query OK, 0 rows affected (0.04 sec)

mysql> INSERT INTO Loans (loan_id, member_id, book_id, loan_date) VALUES
-> (1, 101, 1, '2025-01-05'),
-> (2, 102, 2, '2025-01-08'),
-> (3, 103, 3, '2025-01-10'),
-> (4, 101, 2, '2025-02-01'),
-> (5, 102, 1, '2025-02-05'),
-> (6, 103, 2, '2025-02-12'),
-> (7, 101, 3, '2025-03-01'),
-> (8, 102, 3, '2025-03-07'),
-> (9, 103, 1, '2025-03-15'),
-> (10, 101, 1, '2025-04-01');
ERROR 1452 (23000): Cannot add or update a child row: a foreign key constraint fails ('bookflow_db`.`loans`, CONSTRAINT
`loans_ibfk_1` FOREIGN KEY (`member_id`) REFERENCES `members` (`member_id`))
mysql> INSERT INTO Members (member_id, full_name, email) VALUES
-> (101, 'John Smith', 'john.smith@email.com'),
-> (102, 'Emma Wilson', 'emma.wilson@email.com'),
-> (103, 'Michael Brown', 'michael.brown@email.com');
```

```
mysql> SELECT * FROM books;
+-----+-----+-----+-----+
| book_id | title                               | isbn                | published_year |
+-----+-----+-----+-----+
| 1       | Naruto: The Official Handbook      | 9784088736211      | 2003           |
| 2       | Attack on Titan Guidebook          | 9784063842760      | 2009           |
| 3       | One Piece Data Book                | 9784088725093      | 1997           |
| 4       | Animal Farm                        | 9780451526342      | 1945           |
+-----+-----+-----+-----+
4 rows in set (0.00 sec)

mysql> SELECT * FROM members;
+-----+-----+-----+
| member_id | full_name       | email                |
+-----+-----+-----+
| 1         | Naruto Uzumaki  | naruto@example.com   |
| 2         | Mikasa Ackerman | mikasa@example.com   |
| 3         | Monkey D. Luffy | luffy@example.com    |
| 101      | John Smith      | john.smith@email.com |
| 102      | Emma Wilson     | emma.wilson@email.com |
| 103      | Michael Brown   | michael.brown@email.com |
+-----+-----+-----+
6 rows in set (0.00 sec)
```

```
mysql> SELECT * FROM loans;
+-----+-----+-----+-----+
| loan_id | member_id | book_id | loan_date |
+-----+-----+-----+-----+
| 1       | 101      | 1       | 2025-01-05 |
| 2       | 102      | 2       | 2025-01-08 |
| 3       | 103      | 3       | 2025-01-10 |
| 4       | 101      | 2       | 2025-02-01 |
| 5       | 102      | 1       | 2025-02-05 |
| 6       | 103      | 2       | 2025-02-12 |
| 7       | 101      | 3       | 2025-03-01 |
| 8       | 102      | 3       | 2025-03-07 |
| 9       | 103      | 1       | 2025-03-15 |
| 10      | 101      | 1       | 2025-04-01 |
+-----+-----+-----+-----+
10 rows in set (0.00 sec)

mysql> SELECT * FROM donation_history;
+-----+-----+-----+-----+
| donation_id | book_id | donor_name | donation_date |
+-----+-----+-----+-----+
| 1           | 4       | Raj Kumar  | 2025-07-14    |
+-----+-----+-----+-----+
1 row in set (0.00 sec)
```

AIRLINE MANAGEMENT:

```
mysql> INSERT INTO Passengers (passenger_id, passenger_name, email) VALUES
-> (201, 'Amit Sharma', 'amit.sharma@email.com'),
-> (202, 'Priya Rao', 'priya.rao@email.com'),
-> (203, 'Rahul Verma', 'rahul.verma@email.com'),
-> (204, 'Sneha Kapoor', 'sneha.kapoor@email.com'),
-> (205, 'Karan Mehta', 'karan.mehta@email.com'),
-> (206, 'Neha Singh', 'neha.singh@email.com'),
-> (207, 'Vikram Das', 'vikram.das@email.com'),
-> (208, 'Anjali Nair', 'anjali.nair@email.com'),
-> (209, 'Rohit Gupta', 'rohit.gupta@email.com'),
-> (210, 'Meera Iyer', 'meera.iyer@email.com');
Query OK, 10 rows affected (0.01 sec)
Records: 10 Duplicates: 0 Warnings: 0

mysql> INSERT INTO Bookings (booking_id, passenger_id, flight_id) VALUES
-> (1, 201, 1), (2, 202, 2), (3, 203, 3), (4, 204, 4), (5, 205, 5),
-> (6, 206, 6), (7, 207, 7), (8, 208, 8), (9, 209, 9), (10, 210, 10);
Query OK, 10 rows affected (0.01 sec)
Records: 10 Duplicates: 0 Warnings: 0

mysql> INSERT INTO Flight_History (history_id, flight_id, entry_date) VALUES
-> (1, 1, '2025-01-10');
Query OK, 1 row affected (0.01 sec)
```

```
mysql> CREATE TABLE Flights (flight_id INT PRIMARY KEY, flight_number VARCHAR(20) UNIQUE, source VARCHAR(50), destination VARCHAR(50), departure_date DATE, ticket_price DECIMAL(10,2) CHECK (ticket_price > 0));
Query OK, 0 rows affected (0.03 sec)

mysql> CREATE TABLE Passengers (passenger_id INT PRIMARY KEY, passenger_name VARCHAR(100), email VARCHAR(100) UNIQUE);
Query OK, 0 rows affected (0.03 sec)

mysql> CREATE TABLE Bookings (booking_id INT PRIMARY KEY, passenger_id INT, flight_id INT, FOREIGN KEY (passenger_id) REFERENCES Passengers(passenger_id), FOREIGN KEY (flight_id) REFERENCES Flights(flight_id));
Query OK, 0 rows affected (0.03 sec)

mysql> CREATE TABLE Flight_History (history_id INT PRIMARY KEY, flight_id INT, entry_date DATE, FOREIGN KEY (flight_id) REFERENCES Flights(flight_id));
Query OK, 0 rows affected (0.04 sec)

mysql> INSERT INTO Flights (flight_id, flight_number, source, destination, departure_date, ticket_price) VALUES
-> (1, 'SK101', 'Hyderabad', 'Delhi', '2025-01-10', 4500.00),
-> (2, 'SK102', 'Delhi', 'Mumbai', '2025-01-12', 3800.00),
-> (3, 'SK103', 'Mumbai', 'Chennai', '2025-01-15', 4200.00),
-> (4, 'SK104', 'Chennai', 'Kolkata', '2025-01-18', 5000.00),
-> (5, 'SK105', 'Hyderabad', 'Bangalore', '2025-01-20', 3600.00),
-> (6, 'SK106', 'Bangalore', 'Delhi', '2025-01-22', 4000.00),
-> (7, 'SK107', 'Delhi', 'Hyderabad', '2025-01-25', 4500.00),
-> (8, 'SK108', 'Mumbai', 'Hyderabad', '2025-01-28', 3900.00),
-> (9, 'SK109', 'Kolkata', 'Delhi', '2025-01-30', 4700.00),
-> (10, 'SK110', 'Chennai', 'Mumbai', '2025-02-02', 4300.00);
Query OK, 10 rows affected (0.01 sec)
Records: 10 Duplicates: 0 Warnings: 0
```

```
mysql> INSERT INTO Passengers (passenger_id, passenger_name, email) VALUES
-> (201, 'Amit Sharma', 'amit.sharma@email.com'),
-> (202, 'Priya Rao', 'priya.rao@email.com'),
-> (203, 'Rahul Verma', 'rahul.verma@email.com'),
-> (204, 'Sneha Kapoor', 'sneha.kapoor@email.com'),
-> (205, 'Karan Mehta', 'karan.mehta@email.com'),
-> (206, 'Neha Singh', 'neha.singh@email.com'),
-> (207, 'Vikram Das', 'vikram.das@email.com'),
-> (208, 'Anjali Nair', 'anjali.nair@email.com'),
-> (209, 'Rohit Gupta', 'rohit.gupta@email.com'),
-> (210, 'Meera Iyer', 'meera.iyer@email.com');
Query OK, 10 rows affected (0.01 sec)
Records: 10 Duplicates: 0 Warnings: 0

mysql> INSERT INTO Bookings (booking_id, passenger_id, flight_id) VALUES
-> (1, 201, 1), (2, 202, 2), (3, 203, 3), (4, 204, 4), (5, 205, 5),
-> (6, 206, 6), (7, 207, 7), (8, 208, 8), (9, 209, 9), (10, 210, 10);
Query OK, 10 rows affected (0.01 sec)
Records: 10 Duplicates: 0 Warnings: 0

mysql> INSERT INTO Flight_History (history_id, flight_id, entry_date) VALUES
-> (1, 1, '2025-01-10');
Query OK, 1 row affected (0.01 sec)
```

```
mysql> SELECT * FROM bookings;
```

booking_id	passenger_id	flight_id
1	201	1
2	202	2
3	203	3
4	204	4
5	205	5
6	206	6
7	207	7
8	208	8
9	209	9
10	210	10

```
10 rows in set (0.00 sec)
```

```
mysql> SELECT * FROM flight_history;
```

history_id	flight_id	entry_date
1	1	2025-01-10

```
1 row in set (0.00 sec)
```

flight_id	flight_number	source	destination	departure_date	ticket_price
1	SK101	Hyderabad	Delhi	2025-01-10	4500.00
2	SK102	Delhi	Mumbai	2025-01-12	3800.00
3	SK103	Mumbai	Chennai	2025-01-15	4200.00
4	SK104	Chennai	Kolkata	2025-01-18	5000.00
5	SK105	Hyderabad	Bangalore	2025-01-20	3600.00
6	SK106	Bangalore	Delhi	2025-01-22	4000.00
7	SK107	Delhi	Hyderabad	2025-01-25	4500.00
8	SK108	Mumbai	Hyderabad	2025-01-28	3900.00
9	SK109	Kolkata	Delhi	2025-01-30	4700.00
10	SK110	Chennai	Mumbai	2025-02-02	4300.00

```
10 rows in set (0.00 sec)
```

```
mysql> SELECT * FROM passengers;
```

passenger_id	passenger_name	email
201	Amit Sharma	amit.sharma@email.com
202	Priya Rao	priya.rao@email.com
203	Rahul Verma	rahul.verma@email.com
204	Sneha Kapoor	sneha.kapoor@email.com
205	Karan Mehta	karan.mehta@email.com
206	Neha Singh	neha.singh@email.com
207	Vikram Das	vikram.das@email.com
208	Anjali Nair	anjali.nair@email.com
209	Rohit Gupta	rohit.gupta@email.com
210	Meera Iyer	meera.iyer@email.com

HOSPITAL MANAGEMENT:

```
mysql> CREATE TABLE Doctors (doctor_id INT PRIMARY KEY, doctor_name VARCHAR(100), specialization VARCHAR(50), consultation_fee DECIMAL(10,2) CHECK (consultation_fee > 0));
Query OK, 0 rows affected (0.03 sec)

mysql> CREATE TABLE Patients (patient_id INT PRIMARY KEY, patient_name VARCHAR(100), email VARCHAR(100) UNIQUE);
Query OK, 0 rows affected (0.02 sec)

mysql> CREATE TABLE Appointments (appointment_id INT PRIMARY KEY, doctor_id INT, patient_id INT, appointment_date DATE, FOREIGN KEY (doctor_id) REFERENCES Doctors(doctor_id), FOREIGN KEY (patient_id) REFERENCES Patients(patient_id));
Query OK, 0 rows affected (0.03 sec)

mysql> CREATE TABLE Doctor_History (history_id INT PRIMARY KEY, doctor_id INT, entry_date DATE, FOREIGN KEY (doctor_id) REFERENCES Doctors(doctor_id));
Query OK, 0 rows affected (0.03 sec)

mysql> INSERT INTO Doctors (doctor_id, doctor_name, specialization, consultation_fee) VALUES
-> (301, 'Dr. Arjun Rao', 'Cardiology', 1500.00),
-> (302, 'Dr. Meena Iyer', 'Neurology', 2000.00),
-> (303, 'Dr. Sanjay Patel', 'Orthopedics', 1200.00),
-> (304, 'Dr. Kavita Sharma', 'Dermatology', 1000.00),
-> (305, 'Dr. Ramesh Gupta', 'Pediatrics', 800.00),
-> (306, 'Dr. Anjali Nair', 'Gynecology', 1300.00),
-> (307, 'Dr. Vikram Singh', 'ENT', 900.00),
-> (308, 'Dr. Sneha Kapoor', 'Psychiatry', 1800.00),
-> (309, 'Dr. Rohit Mehta', 'General Medicine', 700.00),
-> (310, 'Dr. Neha Verma', 'Oncology', 2500.00);
Query OK, 10 rows affected (0.01 sec)
Records: 10 Duplicates: 0 Warnings: 0
```

```
mysql> INSERT INTO Patients (patient_id, patient_name, email) VALUES
-> (401, 'Suresh Kumar', 'suresh.kumar@email.com'),
-> (402, 'Anita Sharma', 'anita.sharma@email.com'),
-> (403, 'Rajesh Verma', 'rajesh.verma@email.com'),
-> (404, 'Pooja Singh', 'pooja.singh@email.com'),
-> (405, 'Vikas Das', 'vikas.das@email.com'),
-> (406, 'Ritu Kapoor', 'ritu.kapoor@email.com'),
-> (407, 'Manoj Iyer', 'manoj.iyer@email.com'),
-> (408, 'Deepa Nair', 'deepa.nair@email.com'),
-> (409, 'Kiran Gupta', 'kiran.gupta@email.com'),
-> (410, 'Alok Mehta', 'alok.mehta@email.com');
Query OK, 10 rows affected (0.01 sec)
Records: 10 Duplicates: 0 Warnings: 0

mysql> INSERT INTO Appointments (appointment_id, doctor_id, patient_id, appointment_date) VALUES
-> (1, 301, 401, '2025-01-10'),
-> (2, 302, 402, '2025-01-12'),
-> (3, 303, 403, '2025-01-14'),
-> (4, 304, 404, '2025-01-16'),
-> (5, 305, 405, '2025-01-18'),
-> (6, 306, 406, '2025-01-20'),
-> (7, 307, 407, '2025-01-22'),
-> (8, 308, 408, '2025-01-24'),
-> (9, 309, 409, '2025-01-26'),
-> (10, 310, 410, '2025-01-28');
Query OK, 10 rows affected (0.01 sec)
Records: 10 Duplicates: 0 Warnings: 0
```

```
mysql> SELECT * FROM doctors;
+-----+-----+-----+-----+
| doctor_id | doctor_name | specialization | consultation_fee |
+-----+-----+-----+-----+
| 301 | Dr. Arjun Rao | Cardiology | 1500.00 |
| 302 | Dr. Meena Iyer | Neurology | 2000.00 |
| 303 | Dr. Sanjay Patel | Orthopedics | 1200.00 |
| 304 | Dr. Kavita Sharma | Dermatology | 1000.00 |
| 305 | Dr. Ramesh Gupta | Pediatrics | 800.00 |
| 306 | Dr. Anjali Nair | Gynecology | 1300.00 |
| 307 | Dr. Vikram Singh | ENT | 900.00 |
| 308 | Dr. Sneha Kapoor | Psychiatry | 1800.00 |
| 309 | Dr. Rohit Mehta | General Medicine | 700.00 |
| 310 | Dr. Neha Verma | Oncology | 2500.00 |
+-----+-----+-----+-----+
10 rows in set (0.00 sec)

mysql> SELECT * FROM patients;
+-----+-----+-----+
| patient_id | patient_name | email |
+-----+-----+-----+
| 401 | Suresh Kumar | suresh.kumar@email.com |
| 402 | Anita Sharma | anita.sharma@email.com |
| 403 | Rajesh Verma | rajesh.verma@email.com |
| 404 | Pooja Singh | pooja.singh@email.com |
| 405 | Vikas Das | vikas.das@email.com |
| 406 | Ritu Kapoor | ritu.kapoor@email.com |
| 407 | Manoj Iyer | manoj.iyer@email.com |
| 408 | Deepa Nair | deepa.nair@email.com |
+-----+-----+-----+
```

```
mysql> SELECT * FROM appointments;
```

appointment_id	doctor_id	patient_id	appointment_date
1	301	401	2025-01-10
2	302	402	2025-01-12
3	303	403	2025-01-14
4	304	404	2025-01-16
5	305	405	2025-01-18
6	306	406	2025-01-20
7	307	407	2025-01-22
8	308	408	2025-01-24
9	309	409	2025-01-26
10	310	410	2025-01-28

```
10 rows in set (0.00 sec)
```

```
mysql> SELECT * FROM doctor_history;
```

history_id	doctor_id	entry_date
1	301	2025-01-10

```
1 row in set (0.00 sec)
```